

Data Center Impact Explorer

Team JAMA

Data Science Capstone Project
Data Acquisition and Pre-Processing Report

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1. Identifying Data

1.1 Data Sources:

In our project, we aim to analyze the relationship between data centers and the surrounding communities and investigate the potential impacts of these industrial sites on local areas. As such, we are utilizing multiple data sources on different topics, as we will describe below.

1.1.1 Data on Data Centers

Our central dataset for this project is the FracTracker Alliance U.S. Data Centers Tracker (FracTracker, 2026). This dataset contains over 1,500 individual data centers in the U.S. with their precision locations and additional information such as the operator (e.g., Amazon or Google) and status (e.g., whether the data center is already operational, under construction, or just a proposal). The data is allowed to be downloaded, shared, and used as long as proper credit is given to FracTracker. While many other datasets we identified contain a larger number of data centers, they require payment or subscription to access; so, the open-access FracTracker data was our best option.

In order to investigate the potential impacts of data centers across various axes, a significant component of our project involves the joining of the aforementioned data center dataset with other datasets on different topics based on location. We have identified several datasets pertaining to the environment, electricity production, health, and politics of locations within the U.S., which we will describe in more detail below. While these datasets generally portray past information, and thus cannot be directly related to the proposed, under construction, or only recently operational data centers within the FracTracker dataset, they allow us to learn more about the existing conditions in areas where companies tend to locate their data centers and issues that data center operation has the potential to exacerbate.

1.1.2 Data on Drought

In order to study the potential exacerbation of existing environmental issues due to data centers, we specifically are looking at the levels of drought across the United States. The U.S. Government, specifically the National Drought Mitigation Center at the University of Nebraska-Lincoln, the National Oceanic and Atmospheric Administration, and the U.S. Department of Agriculture, monitor drought in the U.S. and release data every Thursday on the U.S. Drought Monitor (USDM) website (U.S. Drought Monitor, 2026). Drought on the USDM is classified into six different categories, which include normal conditions, abnormally dry (D0), moderate (D1), severe (D2), extreme (D3), and exceptional (D4) drought. The data was downloaded in a CSV format and was filtered by data from the past 20 years (2006 - 2026), which resulted in over 3 million rows of data.

Because this dataset is very detailed and consistent, it is a good source to track environmental changes over time in terms of drought. While these trends in drought conditions could be attributable to a wide variety of causes, it would be insightful to map where data centers are located in relation to areas of severe drought and where their operation may thus pose a particular burden on local water supplies.

1.1.3 Data on Electricity

The electricity component of this project is based on publicly available data from the U.S. Energy Information Administration (EIA) Bulk Data repository. The EIA provides extensive information on electricity generation, customer counts, retail electricity prices, and other energy-related statistics across the United States. We selected this dataset because electricity demand is one of the primary concerns associated with the rapid growth of data centers, particularly as artificial intelligence and cloud computing continue to increase power consumption nationwide.

The original dataset was distributed as approximately 8 GB of concatenated JSON objects, with each record representing a specific combination of geographic region, customer sector, measurement type, and reporting frequency. These included annual, quarterly, and monthly statistics for residential, commercial, industrial, transportation, and all-sector electricity customers, electricity generation, and retail electricity prices. Because the data are reported at the state level, they provide an opportunity to compare trends in electricity production and pricing with the geographic distribution of data centers across the United States. While state-level data cannot directly capture localized impacts around individual facilities, they provide an important baseline for understanding broader regional infrastructure and energy trends relevant to data center development.

1.1.4 Data on Health

Given the relative novelty of the data center boom, longitudinal and/or correlational data specifically focused on how they impact surrounding communities is not yet readily available. However, emerging research is attempting to draw connections between the recognized effects data centers have on the environment and the health risks associated with such ecological shifts. For example, it is widely acknowledged that data centers (especially those that are gas-powered) emit pollutants such as nitrogen oxide, methane, and fine particulate matter, among other compounds; in turn, these very same pollutants are known to increase the frequency of cardiovascular and respiratory conditions (such as asthma or COPD) within a population. Noise pollution and a related decrease in sleep quality are also among emerging concerns (Pavlinich, 2026). Furthermore, some models predict that the ever-increasing demand for data centers and their associated AI computing technologies will push the annual U.S. public health burden in relation to such diagnoses to upwards of \$20 billion by 2030 (Han et al., 2026). It indeed grows increasingly apparent that health concerns must also be considered when discussing the impact of data centers—to this end, our aim for this section is to illustrate how their presence may exacerbate the conditions already affecting a given populace.

To this end, two datasets were selected for investigation. We first intend to utilize the America's Health Rankings (AHR) 2025 Annual Report data provided by the United Health Foundation. This report synthesizes data from over 50 reputable, publicly available data sources and includes over 250 measures related to U.S. public health (United Health Foundation, 2025). While the comprehensiveness of this dataset also means that a significant portion does not pertain to our area of research (for example, data on child welfare or transportation safety), this can easily be remedied with filtering and slicing. Despite the

fact that the report is from 2025, inspection revealed that a vast majority of the data are from 2024; this imbalance, as well as the nature of our second dataset, led us to only retain the 2024 metrics for the sake of consistency. The report and dataset is public access and free to use, with no account required for access or download.

The second dataset to be utilized is the 2024 National Health Interview Survey (NHIS) conducted and published by the National Center for Health Statistics (National Center for Health Statistics, 2024). Specifically, we will be utilizing the data pertaining to adults (individuals over the age of 18), which were largely self-reported in nature with an average response rate of 47.9% after initial contact. All adults included in this dataset are part of the civilian noninstitutionalized population with fixed addresses; unhoused persons, active duty military personnel, persons in care homes, and other individuals in transient living situations were not viable participants. Much like the AHR data, the scope of this survey is quite large—including information about physical and mental health conditions, disabilities, and health care access—and requires substantial modifications to suit our needs. A region (Northeast, Midwest, South, and West), derived from the U.S. Census Bureau, is provided in lieu of county or address in the public dataset download for the sake of participant deidentification (United States Census Bureau). This dataset is also public access and free to use, with the only requirement explicitly stated being proper attribution of credit to the NCHS.

1.1.5 Data on Politics

Another point of interest in this project is the local politics in areas where data centers are situated. Local governments have a large role to play in the construction of data centers in terms of zoning ordinances, permitting, and incentives. As local community members are increasingly mounting opposition to data center construction, the general population's political involvement and leanings are also of interest. As such, we have included several datasets pertaining to local politics. The first is the National Neighborhood Data Archive (NaNDA), which provides voter turnout and partisanship by county from 2004 to 2022 (Clary et al., 2024). This dataset is public access as a downloadable TSV, though requires creating an account or signing in via a university login; redistribution of the data is also not permitted.

Another data source on politics is the American Local Government Elections Database, which contains local election results at the city and county level from 1989 to 2021 (de Benedictis-Kessner et al., 2023). The data is open access under the CC-BY Attribution 4.0 International license. The dataset contains multiple files, but we specifically use `ledb_candidatelevel.csv`, which contains information about individual candidates and election results. While we considered adding another dataset with the county-level voting results for the 2024 federal elections, which is the most recent county-level politics-relevant data available in an aggregate form, we ultimately decided to stick with the above listed datasets as they are far more indicative of the local politicians and political leanings that actually have a bearing on local data center debates.

Finally, one more dataset is required because both political datasets encode location based on FIPS code, which represents a unique county. Since the FracTracker dataset encodes location based on address with ZIP code, we need to convert from one to the other. That is accomplished using a Row Zero dataset of ZIP Codes to County FIPS Codes (Row Zero, 2025). Specifically, in the Row Zero workbook

titled “FIPS code lists and location mapping,” we use the sheet titled “Zip to County.” This provides the FIPS code(s) associated with each ZIP code; as one ZIP code can cross across multiple FIPS codes, it also provides the percentage of the population of each ZIP in the associated FIPS. This dataset is free to use, but requires making a Row Zero account to download.

1.2 Acquisition Process:

The data acquisition process for most of the datasets was a simple download from public sources, sometimes with additional filtering. For the U.S. Drought Monitor source, the data tables were used to filter for data in the past 10 years (2006 - 2026), the “Area Type” was set to County, the “Area” was set for U.S. States and Puerto Rico, and the “Statistics Types” was set to Cumulative Percent Area. For the EIA data, the bulk energy file downloaded and then later parsed during preprocessing. For datasets related to health and politics, download links to the data, codebooks, user guides, and other helpful materials were made readily available on the associated webpages.

The location discrepancy between our two health datasets will likely require additional attention during the integration phase. Since the AHR data includes information at the state level and the NHIS data at the region level, we will have to match each state to a broader region for the sake of effective data communication and utilization in later project phases. This manipulation can only effectively be performed on the AHR data using the region parameters defined in the NHIS methodology; we can easily use state data to create a more general region column, but not the other way around. Luckily, information regarding which states fall under the umbrella of each region is readily available (United States Census Bureau) and can be used to perform the discussed modifications with relative ease.

Another aspect of integrating the data from multiple sources involves joining based on FIPS code, which is how the political and drought datasets encode location information. Since the FracTracker dataset on data centers does not provide FIPS code, we must determine it based on the given address and ZIP code for each listed data center. This is doable using the Row Zero Zip to County dataset (Row Zero, 2025). Sometimes, it is an easy one-to-one match, where a particular ZIP code is enclosed entirely within one FIPS code. However, some ZIP codes cross across multiple FIPS codes; in that case, we use the FIPS code that has the largest proportion of the ZIP code’s resident population. There are about 50 data centers that are missing the ZIP code entirely; in that case, we match to FIPS code based on county and state name. We match on ZIP code rather than county and state name as a default, though, given the issues that could be presented by the slightly different writing of county name across different data sources. This process ultimately allows us to find FIPS code for all but five of the data centers, and thus add FIPS code as an additional column to the FracTracker dataset.

1.3 Issues:

One of the biggest challenges in this project is integrating datasets that were collected by different organizations for different purposes. As mentioned above, different datasets use different geographical units, such as ZIP codes, FIPS codes, counties, states, or regions. Because of this, multiple conversions

and matching steps are required before the datasets can be combined. While these methods allow us to integrate most of the data successfully, there is still the possibility of small matching errors. This situation also precludes us from synthesizing all of our data on different topics, since doing so would erase the geographical granularity that is available for only some of the data. Instead, we will conduct analyses separately and bring together the results later in our final product.

Another issue is that many of the supporting datasets describe historical conditions, so it may not be possible to directly measure the impact of newly proposed or recently constructed data centers. Instead, our analysis will focus on identifying patterns and relationships between existing data centers and surrounding environmental, electrical, health, and political conditions, rather than drawing direct causal conclusions.

2. Data Processing

For clarity, we have split this section based on the processing steps that must be performed for the datasets on separate topics within our project, as each has its own specific processing considerations. We will discuss how we address data cleaning issues such as missing data and interoperability throughout the sections below.

2.1 Data Centers Data Processing

The FracTracker data center dataset contains 1,593 rows, each representing a unique data center. There are ostensibly 44 columns, nine of which pertain to location. Happily, there are few nulls in the location data, which is important because we are joining datasets based on location. Furthermore, even if e.g. the ZIP code is missing (as is the case for 52 rows), location information can still be gleaned from the county and state name.

Out of the remaining 35 columns, 14 contain information that is not particularly relevant for our purposes, such as eight separate columns for information source. So, we can omit them from processing consideration. Unfortunately, out of the remaining columns, 13 of them contain more than 1,000 nulls, which is far too much missing data to impute; so, we must omit these features from our consideration as well. These columns contain information that would have been nice to know, but is not strictly necessary for our purposes, such as power source, number of buildings, and resistance status.

This leaves us with eight potentially useful non-location-related columns: status, operator name, megawattage, sizerank, facility size in square feet, property size in acres, community pushback, and other information. Some of these columns require additional pre-processing for usability; for instance, the facility size is a string with commas to indicate thousands, so those must be stripped out with Regexp before converting the column to numeric.

As mentioned in section 1.2, some additional processing of the FracTracker data center data is required to find the FIPS code associated with each location to enable interoperability with some other

datasets. To join with the additional datasets mentioned below, aggregation based on location is also necessary.

2.2 Environmental Data Processing

The U.S. Drought Monitor is a well-maintained dataset that is regularly updated. This dataset required little pre-processing, since there were no nulls and there is little noise in the data overall. The dataset needed to be cleaned to be able to be used along the other datasets in this project, for example the FracTrackers data. In the USDM dataset, there are 52 states listed because D.C. and Puerto Rico are counted as states. In order to remain consistent with the other datasets, these two states were removed from the dataset since there are little to no data centers in these locations to compare against.

In addition, some clean up of feature types had to be done. The ZIP codes had to be matched with FIPS codes, in a similar process as developed for the political data (see sections 1.1.5, 1.2, and 2.5). The FIPS codes were updated to string types to ease comparison between the ZIP and FIPS datasets. The dates, specifically ValidStart and ValidEnd, were updated to datetime feature types to ease in date comparison.

Finally, the names of the categories of drought are not straightforward to the average reader that is not familiar with this dataset. The drought categories were updated from D0, D1, D2, D3 and D4, to abnormally dry, moderate drought, severe drought, extreme drought, and exceptional drought. This eases the understanding of the labels in future graphs.

2.3 Electricity Data Processing

The EIA electricity data required substantial preprocessing before it could be analyzed. The downloaded bulk file consisted of hundreds of individual JSON objects concatenated together rather than a single structured JSON document. A custom parsing script was developed to separate these objects, extract relevant fields, and organize the information into structured pandas DataFrames.

Each record was categorized according to the type of measurement (customer counts, electricity generation, or retail electricity prices) and reporting frequency (annual, quarterly, or monthly). Annual, quarterly, and monthly datasets were processed separately because they contain different temporal resolutions and numbers of observations. The time-series values stored within each JSON object were expanded so that each year or reporting period became its own column, allowing for straightforward analysis and visualization. Additional preprocessing standardized sector labels by converting abbreviated codes (such as RES, COM, and ALL) into descriptive names, retained geographical identifiers such as state abbreviations and ISO-3166 codes, and converted numeric values into appropriate data types. Only variables relevant to this project were retained in the final datasets to limit the size of the data and to simplify downstream analyses.

Following preprocessing, exploratory data analysis was performed to examine long-term trends in electricity generation and retail electricity prices across states. Initial analyses investigated relationships

between electricity production and the number of operating data centers, as well as changes in industrial electricity prices over time in states with substantial data center activity. These analyses provide a foundation for later integrating electricity infrastructure with the FracTracker data center dataset.

2.4 Health Data Processing

As briefly mentioned in section 1.1.4, both datasets used to investigate health were extremely comprehensive and, as a result, required considerable preprocessing prior to analysis. The National Health Interview Survey data required the most attention of the two, having 630 columns and an unintuitive naming scheme for the included variables. To both facilitate a more informed understanding of the contents and easily “trim the fat” of columns that were completely unnecessary, the dataset’s codebook (National Center for Health Statistics, 2024) was heavily referenced. If a variable was related to a condition or diagnosis corroborated by emerging research related to data centers and health (for example, a condition that is respiratory in nature, as also discussed in section 1.1.4) it was recorded in a spreadsheet along with the associated description.

Variable selection was extremely deliberate for the sake of clarity during later phases; only one variable per condition was selected, and half of the 10 variables ultimately selected utilized a consistent wording schema (for instance, “have you ever had asthma”, “have you ever had cancer”, and “have you ever had COPD”) to further eliminate the potential for ambiguity or confusion. The remaining variables cover sleep quality, weakened immune systems, and symptoms related to year-round (also known as respiratory or environmental) allergies. Although a variable related to a clinical diagnosis of year-round allergies does exist in the dataset and was initially selected for use, it had an unusually high (~50%) percentage of null values. This specific variable would have required complex imputation methods to be usable, but neglecting respiratory allergies in our analyses would mean presenting data that is not wholly representative of the population; since individuals who experience such symptoms far outnumber asthmatic folks, representing these individuals was deemed crucial regardless of a lack of an official diagnosis (Pate et al., 2025).

The America’s Health Rankings data came with its own struggles despite being not nearly as complicated with regards to column quantity (totalling only 10). We could easily remove columns related to confidence interval, value, and score as the primary interest here is the state rank, measure, and year the data were from. Additionally, data where the associated states were “DC” or “ALL” were removed, as they consistently resulted in null rank values. Further inspection showed that, as noted previously, a large portion of the data were from 2024. To maintain alignment with our other dataset, only entries from this year were kept. The true challenge of this data was the measures column—it originally contained 847 unique values, all of which were different combinations of the actual health condition (such as “cancer”) and every demographic metric recorded in the study (such as race, gender, sexual identity, and income level). We first had to identify relevant health measures with the help of their Report Measures table (United Health Foundation, 2025) and then calculate the average rank for each state between all variants of said measures. The result was a dataframe consisting of the average rank for each state for our three selected measures: asthma, cancer, and cardiovascular disease.

2.5 Political Data Processing

Happily, the Row Zero ZIP to County dataset used for ZIP to FIPS code matching is quite clean, without much pre-processing required except for changing datatypes and minor string manipulation to ensure interoperability with the FracTracker dataset. However, the same cannot be said for the political datasets.

The National Neighborhood Data Archive (NaNDA) contains records on voter turnout and partisanship spanning from 2004 to 2022. We can filter the dataset to only the most recent year recorded, which is 2022; there are already features for partisan index based on trends over the preceding six years (i.e., 2016-2022), so there is no need to attempt to aggregate that data over time ourselves. Specifically, there are six columns of interest in this dataset: the FIPS code, number of registered voters, voter turnout percentage, registered voter turnout percentage, partisan index democratic, and partisan index republican. In 2022, this data is very complete, and each column has 3,108 non-null entries. This seems a fairly comprehensive list, given that the U.S. has over 3,000 counties and county-equivalents (Federal Communications Commission, n.d.).

Merging the NANA data with the FracTracker data center data is fairly straightforward after a few key steps. We can group the data center data by FIPS code to find the number of data centers per county and any other information of interest. We also rename the FIPS code column in the NANA data to match FracTracker for ease of merging. Finally, we perform an outer merge between the two datasets to preserve any FIPS codes for data centers that are missing NANA information, and maintain all the FIPS codes from the NANA data that do not have any data centers in them; we can set the data center count to 0 for those columns.

The American Local Government Elections Database requires the most significant pre-processing in this section. The dataset contains 126,599 rows, each of which represents a particular candidate in a particular city- or county-level election between 1989 and 2021. An easy first pre-processing step is to filter to only those candidates who won their elections, since we are interested only in those who actually hold office in these locations.

Based on inspecting the full dataset, the FIPS code is recorded at times inconsistently; while it is often in the correct five-digit format, the FIPS code column sometimes contains extra digits for no clear reason or is written out in text (e.g., “GA-Atlanta”). At the moment, we coerce the column to numeric and drop the nulls that result; another option would be to use the listed state and county to attempt to find the FIPS code, which we may later pursue if we find the remaining data to be insufficient.

Since we are interested in who may be in power in cities and counties where data centers are currently proposed, under construction, or operating, we aim to understand recent trends in the election of officials. As such, we filter the dataset to only candidates who were elected in 2016 or later, which leaves 15,259 rows in the data.

The full American Local Government Elections Database contains 32 columns, but we keep only nine that are relevant to our task: FIPS code, county name, state abbreviation, type of political office, year, win or lose, probability that the politician is a democrat, probability that they are a democrat, and resulting estimated political affiliation. After performing an outer merge with the FracTracker data based

on FIPS code, there are no or very few nulls in any columns except for probability Democrat, probability Republican, and political affiliation, which each have 893 nulls. We must filter out those nulls as well since we are mainly interested in the political affiliation of those in power and do not have a way to impute, leaving us with a final count of 14,366 rows covering 1,553 unique FIPS codes.

We will further progress the analysis of this merged dataset by collapsing the trends per FIPS code: finding the proportion of Democratic/Republican elected officials between 2016 and 2021, and determining if the trend is towards a particular party gaining more power.

3. Appendix

All of our code for this project is available on our [group GitHub](#), which we will continue to update as the work proceeds. Below, we will provide links to the code written specifically for different aspects of this project, provide sample data from the various cleaned datasets, and define the columns.

3.1 Data Centers Code and Sample Data

Initial inspection and cleaning of the FrackTracker data center data (FrackTracker, 2026), prior to its merge with any other datasets, can be found in [Inspection Data Centers and Politics.ipynb on GitHub](#). Below, we present a sample of the data with the most relevant features to our interests.

	facility_name	city	state	zip	county	status	operator_name	mw	sizerank	facility_size_sqft	property_size_acres
2	Prudhoe Bay Data Center	Prudhoe Bay	AK	99734.0	North Slope	Operating	Far North Digital, LLC	120.0	Hyperscale (100-999 MW)	NaN	100.0
3	Grant County Data Center	Sheridan	AK	72150.0	Grant	Proposed	Clean Cloud Energy	500.0	Hyperscale (100-999 MW)	1000000.0	753.0
4	Project Marvel	Bessemer	AL	35022.0	Jefferson	Proposed	Logistic Land Investments LLC	1200.0	Mega campus (> 1,000 MW)	4500000.0	1600.0
5	BHM01 Nebius Data Center	Birmingham	AL	35221.0	Jefferson	Proposed	Nebius	300.0	Hyperscale (100-999 MW)	NaN	80.0
6	DC Blox	Birmingham	AL	35233.0	Jefferson	Operating	NaN	NaN	Unknown	NaN	NaN

The columns are defined as follows:

- **facility_name**: Name of the data center facility
- **city**: City in which data center is located
- **state**: State in which data center is located
- **zip**: ZIP code in which data center is located
- **county**: County in which data center is located
- **status**: Operating status of the data center (e.g., proposed, under construction, operating)
- **operator_name**: Developer or primary operator for the data center
- **mw**: Power capacity of the data center in megawatts
- **sizerank**: Categorical size classification based on mw column
- **facility_size_sqft**: Building footprint in square feet
- **property_size_acres**: Total land area of the property in acres

3.2 Environmental Aspect Code and Sample Data

The figure below is a screenshot of a sample of the U.S. Drought Monitor dataset. The features are explained in detail below.

	MapDate	FIPS	County	State	None	D0	D1	D2	D3	D4	ValidStart	ValidEnd
0	20260630	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-30	2026-07-06
1	20260623	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-23	2026-06-29
2	20260616	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-16	2026-06-22
3	20260609	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-09	2026-06-15
4	20260602	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-02	2026-06-08

- **MapDate**: Date conditions are reported
- **FIPS**: Federal Information Processing Standards, five digit number associated with the location of the drought conditions
- **County**: County location of condition
- **State**: State location of condition (State name abbreviation)
- **None**: Normal or wet condition percentage
- **D0**: Abnormally dry condition percentage
- **D1**: Moderate drought condition percentage
- **D2**: Severe drought condition percentage
- **D3**: Extreme drought condition percentage
- **D4**: Exceptional drought condition percentage
- **ValidStart**: Date of the start of the reported conditions
- **ValidEnd**: Date of the end of the reported conditions

3.3 Electricity Aspect Code and Sample Data

Electricity generation, customer account and electricity price data were obtained from U.S. Energy Information Administration. The original EIA bulk data was parsed and transformed into structured CSV files, with separate datasets created for annual, quarterly and monthly observations. Initial preprocessing and restructuring of these datasets can be found in the preprocessing notebooks on GitHub. The transformed datasets provide a consistent state-level format that simplifies exploratory analysis and comparison across electricity demand, generation, and pricing trends.

series_id	name	geography	iso3166	sector	units	2001
ELEC.PLANT.GEN.63914-ALL-ALL.A	Net generation : Cedar Springs I (63914) : all...	USA-WY	USA-WY	GEN	megawatthours	NaN
ELEC.PLANT.GEN.63915-ALL-ALL.A	Net generation : Cedar Springs II (63915) : al...	USA-WY	USA-WY	GEN	megawatthours	NaN
ELEC.PLANT.GEN.63917-ALL-ALL.A	Net generation : DC Water Solar (63917) : all ...	USA-DC	USA-DC	GEN	megawatthours	NaN
ELEC.PLANT.GEN.63918-ALL-ALL.A	Net generation : POET Biorefining - Hudson (63...	USA-SD	USA-SD	GEN	megawatthours	NaN
ELEC.PLANT.GEN.63820-ALL-ALL.A	Net generation : Huneke CSG 1, LLC (63820) : a...	USA-MN	USA-MN	GEN	megawatthours	NaN

Below, we present the key features shared across the datasets. The columns are defined as follows:

- **series_id**: Unique identifier assigned by the EIA for each time series.
- **name**: Descriptive name of the metric represented by the time series
- **geography**: Geographic region covered by the data (typically a U.S state).
- **iso3166**: ISO3166 geographic code corresponding to the state.
- **sector**: Electricity sector represented by the data (residential, commercial, transportation, or all)
- **units**: Units in which the values are reported (number of customers, megawatthours or cents per kilowatthour)

Three categories of electricity data were used:

- **Electricity Customers (ELEC.CUSTOMERS)**: Number of customer accounts by state and sector
- **Electricity Generation (ELEC.PLANT.GEN)**: Net electricity generation by state and sector
- **Electricity Prices (ELEC.PRICE)**: Average retail electricity prices by state and sector.

For each category, separate datasets were maintained for annual (A), quarterly (Q), and monthly (M) observations to support analyses at different temporal resolutions.

3.4 Health Aspect Code and Sample Data

All code for development of the America's Health Rankings data can be found at [Health_Data_Cleaning_and_Investigation.ipynb](#) under the AHR Data dropdown, in which we perform minor pre-processing steps to select the columns and measures of interest. A snapshot of the cleaned and sliced dataset is presented below.

Report	Measure	State	Rank	Data Year(s)
60526 2025 Annual	Asthma	AK	23.0	2024
60527 2025 Annual	Asthma	AL	26.0	2024
60529 2025 Annual	Asthma	AR	30.0	2024
60530 2025 Annual	Asthma	AZ	30.0	2024
60531 2025 Annual	Asthma	CA	1.0	2024
...	...	...	...	...
64421 2025 Annual	Cardiovascular Diseases	VT	28.0	2024
64422 2025 Annual	Cardiovascular Diseases	WA	3.0	2024
64423 2025 Annual	Cardiovascular Diseases	WI	32.0	2024
64424 2025 Annual	Cardiovascular Diseases	WV	49.0	2024
64425 2025 Annual	Cardiovascular Diseases	WY	16.0	2024

147 rows × 5 columns

The remaining columns are defined as follows:

- **Report**: The report all data in the dataset were published within. All data are from the 2025 Annual report and are not pulled from other previous reports.

- **Measure:** The health condition measures selected based on current and emerging research on the interplay between data centers, their environmental impact, and resulting health problems (see section 1.1.4). Consists of “asthma”, “cardiovascular diseases”, and “cancer”.
- **State:** The U.S. state being measured.
- **Rank:** The average rank for each measure (calculated across all measures) for each state.
- **Data Year(s):** Year the data were initially collected in. All data were collected in 2024.

The code for the development of the National Health Interview Survey can be found in the same [.ipynb](#) file under the NHIS Data dropdown. There, minor preprocessing and column selection is conducted. The head of the cleaned and trimmed dataset is presented below.

	REGION	URBRRRL23	HHX	CURRESP_A	HLTHCOND_A	MEDRXTRT_A	COPDEV_A	CANEV_A	ASEV_A	SLPREST_A	DEPEV_A	ANXEV_A
0	3	4	H067658	2	2	2	2	2	2	3	2	2
1	3	4	H076577	2	2	2	2	2	2	3	2	2
2	3	4	H019335	1	2	2	1	2	2	4	2	2
3	3	4	H012701	1	2	2	2	2	2	2	2	1
4	3	2	H049678	1	2	2	2	2	1	3	1	1

The columns are defined as follows. For the included measures, the question as listed in the codebook (National Center for Health Statistics, 2024) is provided:

- **REGION:** Region the participant is from. The included regions are Northeast (1), Midwest (2), South (3), and West (4), and are divided based on the U.S. Census Bureau Geographic Levels (United States Census Bureau)
- **URBRRRL23:** Urbanization level of where the participant is from. The included levels are large central metro (1), large fringe metro (2), medium and small metro (3), and nonmetropolitan (4) and are determined based on the NCHS Urban Rural Classification Scheme (CDC, 2024).
- **HHX:** Unique household identifier of the participant.
- **CURRESP_A:** “Do you get symptoms such as sneezing, runny nose, or itchy or watery eyes due to hay fever, seasonal or year-round allergies?”
- **HLTHCOND_A:** “Do you currently have a health condition that a doctor or other health professional told you weakens the immune system?”
- **MEDRXTRT_A:** “In the past 12 months, have you taken prescription medication or had any medical treatments that a doctor or other health professional told you would weaken your immune system?”
- **COPDEV_A:** “Have you EVER been told by a doctor or other health professional that you had... Chronic Obstructive Pulmonary Disease, C.O.P.D., emphysema, or chronic bronchitis?”
- **ASEV_A:** “Have you EVER been told by a doctor or other health professional that you had asthma?”
- **SLPREST_A:** “During the past 30 days, how often did you wake up feeling well-rested? Would you say never, some days, most days, or every day?”
- **DEPEV_A:** “Have you EVER been told by a doctor or other health professional that you had... Any type of depression?”
- **ANXEV_A:** “Have you EVER been told by a doctor or other health professional that you had... Any type of anxiety disorder?”

3.5 Political Aspect Code and Sample Data

The code for the development of our strategy to convert from ZIP to FIPS can be found on our GitHub within [Dev ZIP to FIPS.ipynb](#). There, we do minor pre-processing for the Row Zero ZIP to County dataset (Row Zero, 2025). The head of the cleaned dataset is presented below.

	ZIP	County FIPS code	USPS Default State for ZIP	County	County State	State Abbreviation	ZIP Code Population	% of ZIP Residents in County	# of Counties	USPS Default City for ZIP	Ratio_of_ZIP_Res_in_County
0	35004	1115	AL	St. Clair County	Alabama	AL	11,558	100%	1	Moody	1.00
1	35005	1073	AL	Jefferson County	Alabama	AL	8,143	100%	1	Adamsville	1.00
2	35006	1073	AL	Jefferson County	Alabama	AL	3,599	93%	3	Adger	0.93
3	35006	1127	AL	Walker County	Alabama	AL	3,599	5%	3	Adger	0.05
4	35006	1125	AL	Tuscaloosa County	Alabama	AL	3,599	2%	3	Adger	0.02

The columns are defined as follows:

- **ZIP:** ZIP code, which may be repeated in different rows given association with different counties.
- **County FIPS code:** FIPS code associated with a specific county. FIPS code is a five-digit number but may begin with 0, in which case it is written with four digits as a numeric column; this is maintained throughout the other datasets that will be merged, so the necessary consistency is present.
- **USPS Default State for ZIP:** The state associated with the ZIP code as defined by the U.S. Postal Service.
- **County:** County name associated with FIPS code.
- **County State:** State in which county is located.
- **State Abbreviation:** Abbreviation of listed state.
- **ZIP Code Population:** Number of residents residing within ZIP code overall.
- **% of ZIP Residents in County:** Percentage of ZIP code residents that live within the listed county/FIPS code.
- **# of Counties:** Number of counties that this ZIP code crosses over.
- **USPS Default City for ZIP:** Default city that the U.S. Postal Service uses for this ZIP code.
- **Ratio_of_ZIP_Res_in_County:** Ratio of ZIP code residents that live in this county, processed from the % of ZIP Residents in County column.

The code for pre-processing the National Neighborhood Data Archive and American Local Government Elections Database datasets and merging each with the FracTracker data center data can be found on our GitHub within [Preprocessing Politics.ipynb](#). Below, first find a sample from the cleaned and merged National Neighborhood Data Archive dataset (Clary et al., 2024).

	fips	data_center_count	REG_VOTERS	VOTER_TURNOUT_PCT	REG_VOTER_TURNOUT_PCT	PARTISAN_INDEX_DEM	PARTISAN_INDEX_REP
0	1001.0	0.0	43488.0	0.397022	0.410780	0.254452	0.745548
1	1003.0	0.0	189028.0	0.400008	0.389419	0.203950	0.796050
2	1005.0	0.0	17270.0	0.333925	0.381413	0.437367	0.562633
3	1007.0	0.0	14741.0	0.336907	0.400787	0.196562	0.803438
4	1009.0	0.0	41794.0	0.366511	0.399124	0.089329	0.910671

The columns are defined as follows:

- **fips**: FIPS code
- **Data_center_count**: Number of data centers
- **REG_VOTERS**: Number of registered voters
- **VOTER_TURNOUT_PCT**: Percentage of ballots cast out of number of citizens of a voting age
- **REG_VOTER_TURNOUT_PCT**: Percentage of ballots cast out of number of registered voters
- **PARTISAN_INDEX_DEM**: Six-year aggregate Democratic partisanship index, averaging presidential and senate Democratic vote ratios over 2016-2022
- **PARTISAN_INDEX_REP**: Six-year aggregate Republican partisanship index, averaging presidential and senate Republican vote ratios over 2016-2022

Finally, below please find a sample from the cleaned and merged American Local Government Elections Database dataset (de Benedictis-Kessner et. al., 2023).

	fips	data_center_count	geo_name	state_abb	office_consolidated	year	winner	prob_democrat	prob_republican	pid_est
0	1001.0	0.0	autauga	AL	Sheriff	2018	win	0.0	1.0	R
1	1003.0	0.0	baldwin	AL	Sheriff	2018	win	0.0	1.0	R
2	1003.0	0.0	baldwin	AL	County Legislature	2018	win	0.0	1.0	R
3	1003.0	0.0	baldwin	AL	County Legislature	2018	win	0.0	1.0	R
4	1003.0	0.0	baldwin	AL	County Legislature	2018	win	0.0	1.0	R

The columns are defined as follows:

- **fips**: FIPS code
- **Data_center_count**: Number of data centers
- **geo_name**: Name of local government
- **state_abb**: Abbreviation of state
- **office_consolidated**: Office for which candidate ran
- **year**: Year in which election occurred (filtered to 2016-2021)
- **winner**: Whether candidate won or lost election (filtered to only winners)
- **prob_democrat**: Probability that candidate is a Democrat. For partisan elections, this is based on official election returns; for non-partisan elections, the study authors found a probabilistic estimate of whether the candidate is a Democrat.
- **prob_republican**: Probability that candidate is a Republican. For partisan elections, this is based on official election returns; for non-partisan elections, the study authors found a probabilistic estimate of whether the candidate is a Republican.

- *pid_est*: Estimate of best partisan category for the candidate based on prob_democrat and prob_republican.

4. Table of Contributions

The table below identifies contributors to various sections of this document.

	Section	Writing	Editing
1	Data Sources	Andrea MacGregor, Anna Grace, Jillian Kunze, Morgan Murphy	Andrea MacGregor, Anna Grace, Jillian Kunze, Morgan Murphy
2	Data Pre-Processing	Andrea MacGregor, Anna Grace, Jillian Kunze, Morgan Murphy	Andrea MacGregor, Anna Grace, Jillian Kunze, Morgan Murphy
3	Appendix	Andrea MacGregor, Anna Grace, Jillian Kunze, Morgan Murphy	Andrea MacGregor, Anna Grace, Jillian Kunze, Morgan Murphy

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